



FAIR WITNESS

Trust-Minimized Execution for Autonomous Financial Agents

AI proposes. Deterministic policy authorizes. Treasury executes.

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Public-testnet controlled demonstration

The market conditions used in the public demo are synthetic. The cross-chain transactions, Attestcoin proof path, Creditcoin verification, deterministic policy evaluation, replay controls, and bounded treasury execution are real public-testnet operations.

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Executive Snapshot

Layer	Responsibility	Trusted for capital authorization?
Attestcoin	Prove source-chain transaction inclusion/continuity and attested height	Yes - through on-chain verification
AI reasoning	Select EXECUTE or WAIT for a deterministic candidate	No
Treasury policy	Recompute mandate, market, portfolio, replay and amount constraints	Yes
Execution adapter	Enforce the exact venue/pair/recipient path	Yes
Supabase / frontend	Observability and UX	No

1. Executive Summary

Fair Witness is a public-testnet reference system for autonomous financial agents that need to act on cross-chain information without giving an AI model arbitrary transaction authority or custody of user capital.

The architecture separates intelligence from authority. Cross-chain market facts originate on Ethereum Sepolia. Attestcoin provides cryptographically verifiable transaction evidence. Deterministic code derives a strategy candidate. The reasoning layer chooses only EXECUTE or WAIT. A user-owned Creditcoin treasury then verifies the Attestcoin evidence again, recomputes policy from current on-chain state, and either records a typed rejection or executes through a fixed adapter.

Core thesis: an AI agent can contribute timing and judgment without becoming the security boundary.

The model cannot choose arbitrary targets, routes, recipients, calldata, token pairs, venues, or action amounts. This makes model quality primarily a liveness and decision-quality concern rather than a direct capital-safety assumption.

2. The Problem

Most autonomous-agent stacks collapse reasoning and execution into one trust domain: the same process that decides what should happen also holds keys, chooses transaction parameters, and broadcasts calls. In finance, this creates a dangerous equivalence between a recommendation and authority.

A robust design must assume failures across multiple layers: model hallucination, prompt manipulation, stale context, compromised hosts, stolen submitter keys, misleading RPC responses, corrupted databases, and cross-chain data that is cryptographically or semantically invalid.

Fair Witness therefore requires independent answers to three questions:

1. Are the external facts real?
2. Is the proposed action allowed by the user mandate?
3. Can execution occur only through the intended path?

Attestcoin verifies cross-chain evidence.
 Deterministic policy authorizes bounded intent.
 The treasury executes through a constrained adapter.

3. Design Principles

3.1 Reasoning is not authority

The AI may decide whether a deterministic candidate should proceed, but it never defines the transaction program.

3.2 Cross-chain facts must be verifiable

Source-chain values are not trusted merely because an API or agent process reports them. Attestcoin evidence is verified on Creditcoin before it can become authorization input.

3.3 Policy is recomputed independently

The treasury treats agent-side calculations and preflight checks as non-authoritative. It recomputes market, portfolio, amount, replay, rate, and mandate constraints from verified evidence and current chain state.

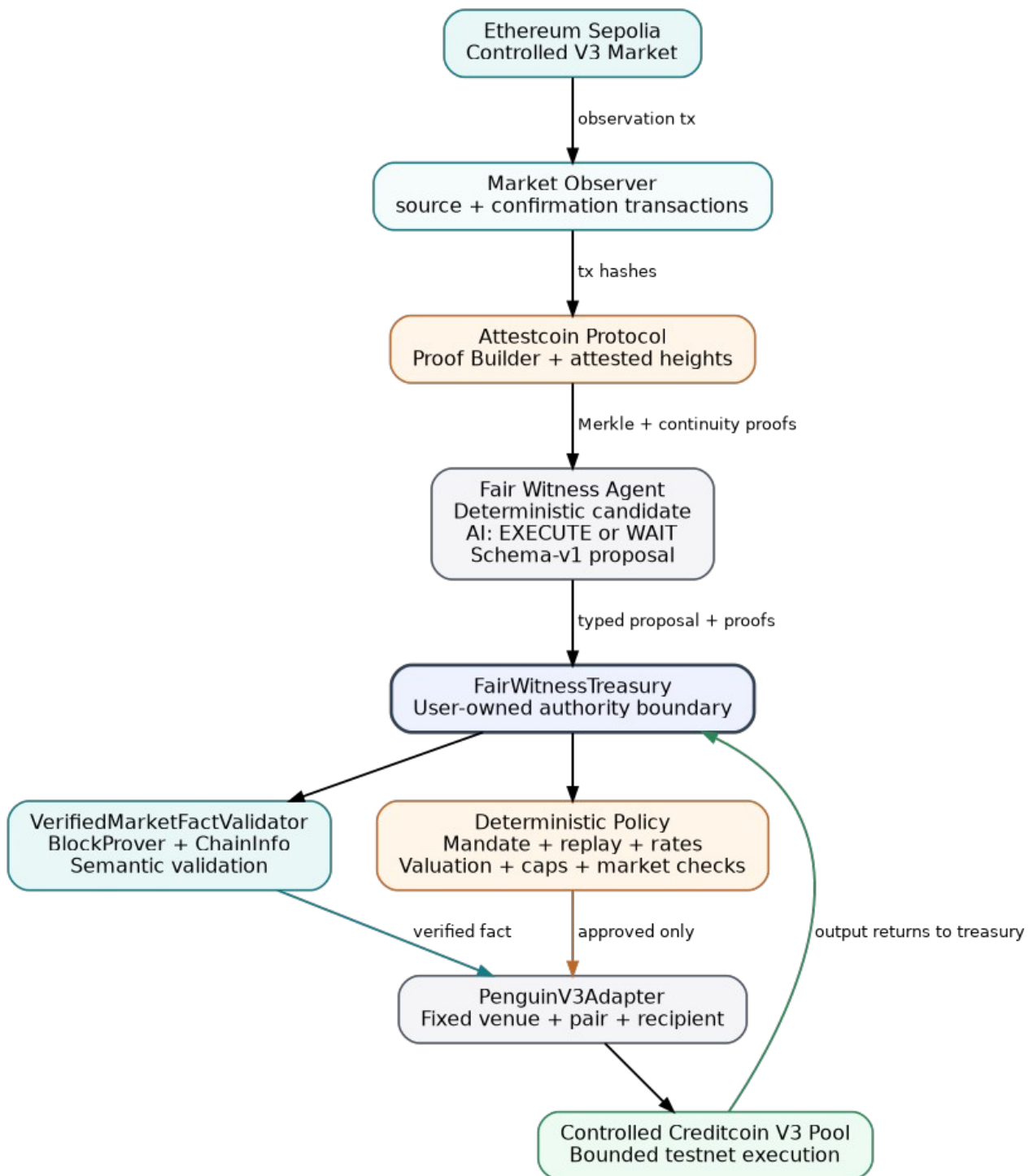
3.4 Execution is deliberately narrow

The adapter is bound to the intended venue and pair, and output returns to the calling treasury. The proposal format has no arbitrary target, selector, recipient, route, or calldata field.

3.5 Auditability must survive model failure

Typed attempts, reason codes, cryptographic commitments, nonces, proposal identities, execution keys, and policy epochs make rejected and successful paths independently inspectable.

4. System Architecture



Each user receives an independent FairWitnessTreasury from a permissionless factory. The treasury stores the immutable mandate, registered submitters, automation state, replay/rate state, and append-only attempt journal. It is the only component that can approve strategy assets for execution.

The current lifecycle factory uses external bytecode stores so it can deploy the full treasury creation code without exceeding the EIP-170 factory runtime-size limit.

Component	Role
FairWitnessTreasury	Custody, deterministic authorization, replay/rate enforcement, attempt journal
FairWitnessTreasuryFactory	Permissionless creation of isolated user treasuries
VerifiedMarketFactValidator	Attestcoin verification + semantic source binding
PenguinV3Adapter	Fixed destination market reads and constrained execution
Agent service	Observation, proof acquisition, deterministic candidate, reasoning, proposal submission
Frontend	Mandate UX, live telemetry, Activity, Decision Detail, Verify, Safeguards
Supabase	Optional projection/observability only; never authorizes capital movement

5. Attestcoin Protocol Integration

Attestcoin is a security-critical dependency in Fair Witness rather than an auxiliary oracle feed. The current source is Ethereum Sepolia (chain ID 11155111, Attestcoin chain key 1) and the destination is Creditcoin testnet (chain ID 102031).

5.1 SDK and proof acquisition

The agent integrates @gluwa/usc-sdk through agent/src/attestcoinClient.ts. It uses the ProofBuilder to obtain transaction proof packages, PrecompileChainInfoProvider to check supported/attested heights, and PrecompileBlockProver for a read-only pre-verification.

```
await proofBuilder.waitForHeightAttested(chainKey, height)
await chainInfoProvider.waitForHeightAttested(chainKey, height)

proof = {
  chainKey, blockHeight, transactionIndex,
  encodedTransaction, merkleProof, continuityProof
}
```

The agent deliberately waits for both the proof-builder service and the on-chain ChainInfo view. A proof-builder cache result is not treated as equivalent to authoritative Creditcoin attestation state.

5.2 Dual-proof freshness model

Fair Witness uses both a source observation and a later confirmation observation. The confirmation must be newer, within a bounded block gap, and recent relative to the current Attestcoin height. This blocks the use of two internally consistent but stale proofs.

5.3 On-chain Creditcoin verification

VerifiedMarketFactValidator performs the authoritative verification through the Creditcoin BlockProver and ChainInfo precompiles, then recomputes the Merkle transaction index and semantically decodes the proven source transaction.

Attestcoin surface	Address / role
BlockProver	0x00FD2 -

Cryptographic inclusion answers whether a transaction was included. Fair Witness must also determine whether that transaction has the application-specific meaning the policy expects. The semantic decoder binds the proven transaction to the immutable observer and source pool and validates the expected receipt/event semantics.

A valid proof of the wrong transaction is not valid authorization.

Attestcoin proves cross-chain inclusion/continuity and exposes attested source-chain state to Creditcoin. It does not by itself prove token equivalence, bridge availability, natural arbitrage, profitability, or future execution price. Fair Witness keeps these boundaries explicit.

```

graph LR
    A[UNTRUSTED  
AI / Prompt / Provider] --> B[Bounded Proposal  
No arbitrary calldata, route,  
recipient, target or amount]
    C[UNTRUSTED  
Agent Host / Submit Key] --> B
    D[UNTRUSTED  
Frontend / Supabase / RPC] -.-> B
    B --> E[Attestcoin + Semantic  
Evidence Verification]
    E --> F[Deterministic Treasury Policy  
Recompute eligibility + caps + replay]
    F -- approved --> G[Constrained Adapter  
Exact pair / venue / recipient]
  
```

AI may	AI may not
Choose EXECUTE or WAIT	Choose arbitrary target contract or selector
Explain its rationale	Provide executable calldata or route
Interpret verified context	Change token pair, venue, recipient or amount
Decline an eligible candidate	Override policy, replay, rate, freshness or liquidity limits

Universal policy checks are combined with a strategy-specific branch. The treasury verifies the active mode and policy hash, source evidence, destination market state, replay identities, rate limits, balances, slippage, direction, and the exact permitted amount before execution.

- enabled strategy bitmap
- maximum action value
- maximum slippage
- source drift and proof freshness
- minimum source and destination liquidity
- maximum destination spot/TWAP deviation
- attempt and execution rate limits
- asset pair and venue equality
- deadline and nonce/replay state
- available treasury balance

Normal registered-agent failures are represented as typed business rejections where possible. A rejected attempt must not create a token approval, mutate successful execution replay state, increment successful execution counters, or move strategy capital.

8. Strategy Model

8.1 Risk Reduction

Highest priority. If WCTC exposure exceeds the immutable maximum, the only valid direction is to reduce WCTC exposure. The value is capped by excess exposure, universal action limit, per-action risk limit, daily risk limit, and available balance.

8.2 Rebalancing

Compares current WCTC allocation against an immutable target and tolerance band. Only movement toward the target is valid, and the input amount is deterministically derived with conservative rounding.

8.3 Arbitrage

Compares Attestcoin-verified source reference state with destination TWAP while enforcing direction, liquidity, source drift, destination manipulation bounds, pool fee, slippage reserve, minimum net edge, and action caps.

8.4 Fixed priority

Risk Reduction -> Rebalancing -> Arbitrage

After a successful execution, lower-priority candidates derived from the old portfolio snapshot are discarded and must be recomputed.

9. Proposal, Replay, and Audit Model

The schema-v1 proposal represents bounded intent rather than a programmable transaction. It commits to strategy/action, asset direction, venue, exact input, slippage ceiling, deadline, nonce, and evidence/observation/decision/policy hashes.

Replay control	Purpose
Attempt ID	Monotonic audit identity for every admitted attempt
Agent nonce	Prevents nonce reuse for processable first submissions
Proposal ID	Detects exact replay
Strategy/evidence execution key	Prevents changed-nonce/agent/amount re-execution of the same fact for a strategy
Policy epoch	Invalidates proposals across pause/resume policy transitions

A model WAIT is an off-chain decision and is not relabeled as an on-chain policy rejection. Activity and Decision Detail reconstruct authoritative on-chain attempts; Verify independently locates an attempt by treasury address and attempt ID.

10. Security Model

Fair Witness explicitly treats the AI, prompt, model provider, agent host, database, frontend, public RPCs, and registered submit key as untrusted or compromiseable for authorization purposes.

Security theorem: even if the reasoning layer produces malicious, incorrect, oversized, stale, replayed, or unauthorized instructions, it must not cause capital movement outside deterministic policy constraints.

10.1 Key invariants

4. Treasury capital never enters the agent EOA.
5. No AI-accessible field can select an arbitrary target, route, recipient, token, selector, or calldata.
6. Source facts are accepted only after Attestcoin proof verification and semantic validation.
7. Rejected attempts create no strategy-token approval or capital movement.
8. The adapter output recipient is always the calling treasury.
9. Policy recomputes direction, valuation, allocation/exposure, amount cap and replay state.
10. A strategy/evidence identity executes at most once per treasury.
11. Supabase loss or compromise cannot move capital or change policy.

10.2 Adversarial test coverage

Case	Expected protection
Oversized action	AmountExceedsPolicy; balances/allowances/execution state unchanged
Unauthorized asset / venue	Rejected before capital path
Stale / invalid evidence	Never accepted as verified authorization
Exact replay	No second execution; replay attempt inspectable
Changed-nonce evidence replay	Strategy-scoped evidence key blocks re-execution

Wrong-direction rebalance	Cannot increase deviation
Risk cap / daily limit	Cannot exceed configured risk reduction budget
Adapter failure	Atomic rollback of capital/replay effects

11. Controlled Public-Testnet Demonstration

Because no sufficiently active comparable market existed across the supported public testnets, Fair Witness uses explicitly controlled V3 test markets. This allows the project to exercise the full cross-chain verification and execution path without misrepresenting synthetic price conditions as naturally occurring economic opportunity.

The market conditions are synthetic; the verification and execution path are not.

11.1 Current lifecycle deployment

Component	Value
Creditcoin chain ID	102031
Factory	0x494490bBF748e59a659227F46510535BF3818442
Factory deployment block	5465730
Controlled faucet / reserve	0x477564A6e66966d2fcb5E3CaE33282e8d25dD71A
Validator	0x13Dd030815550080Ef80Ff3499fAE8d971A119f5
Destination adapter	0x9bAF94da27d5C71c42b40D25b43070083DE7296E
Sepolia observer	0x9bAF94da27d5C71c42b40D25b43070083DE7296E
Sepolia source pool	0xB88deB0436eAD37A6Dd625e9140AE45D5024424f

11.2 Example Attestcoin-backed execution

Evidence	Value
Source observation tx	0x1c0e67ad9621ec5d23f061d330e1b7b41d69c66646a97358e1be09dfe42e408f
Confirmation observation tx	0xf87e562164ca1f23dea01df67b0f907cd36f9e2b25d49f99133a74f43a906cbb
Source / confirmation blocks	11667084 / 11667085
Creditcoin execution tx	0x05b04421472d9318e297b255ad233d974839518e88d6f7ee981d64d4ec3e58a4
Treasury attempt	2
Recorded path	VALID_ATTESTCOIN_RISK_REDUCTION

The controlled fwUSD/fwWCTC assets are independently issued testnet assets. They do not imply a bridge, redemption relationship, economic peg, natural arbitrage, production liquidity, or profitability.

12. Product Experience

12. Sign in with the embedded-account flow.
13. Build an immutable mandate and deploy a user-owned treasury.
14. Fund controlled test assets, authorize the bounded submitter, and enable autonomous mode.
15. Observe real per-treasury telemetry across Observe -> Prove -> Reason -> Authorize -> Execute.
16. Inspect reason-coded attempts in Activity.
17. Open Decision Detail for the complete schema-v1 on-chain attempt.
18. Use Verify to independently locate the same treasury attempt.

The UI distinguishes off-chain WAIT decisions from on-chain policy attempts and does not fabricate model rationale from chain fields. Authority-bearing facts are read from chain; browser validation is UX only.

13. Limitations

- The demo markets are controlled rather than naturally occurring comparable markets.
- Testnet liquidity and oracle history can be thin or manipulable.
- Model/provider availability affects liveness.
- The agent service is centralized for scheduling/liveness.
- A compromised registered submitter can cause bounded gas/storage grief until deregistered.
- Immutable-policy mistakes require migration to a new treasury.
- Controlled source/destination tokens do not establish cross-chain economic identity.
- The demo does not prove an end-to-end profitable two-leg arbitrage cycle.

These limitations constrain claims; they do not justify weakening the capital-safety controls.

14. Roadmap and Creditcoin Ecosystem Opportunity

Fair Witness is designed as a reusable authorization architecture rather than a single trading bot. The same pattern can be applied wherever an autonomous system needs verified external-chain facts and bounded on-chain authority.

Phase 1 - Hardened agent infrastructure

- isolated signer infrastructure
- leader election / duplicate-runner protection
- production monitoring and incident response
- production indexing and reorg handling
- external security review

Phase 2 - Richer verified facts

- additional Attestcoin-supported source chains
- more source observers and semantic decoders
- additional Creditcoin-native market/risk adapters
- reusable verified-fact schemas

Phase 3 - Broader autonomous finance

- treasury management

- lending and risk controls
- collateral monitoring
- RWA servicing triggers
- insurance conditions
- governance automation
- cross-chain credit and portfolio policies

```
verified external fact
+
bounded machine judgment
+
deterministic on-chain authorization
=
safer autonomous execution
```

15. Conclusion

Fair Witness demonstrates a different way to build AI agents on-chain. Instead of asking whether a model is trustworthy enough to hold transaction authority, the system assumes the model can be wrong and designs around that fact.

Attestcoin provides verifiable cross-chain evidence. Deterministic policy defines what is allowed. A constrained treasury and adapter define how execution can occur. The AI contributes judgment without becoming custody, oracle, policy engine, or transaction programmer.

AI proposes. Deterministic policy authorizes. Treasury executes.

Appendix A. Current Deployment

The machine-readable deployment source of truth is `contracts/deployments/controlled-demo-schema-v1-lifecycle.json`. Historical controlled execution/rejection evidence is preserved in `contracts/deployments/controlled-demo-schema-v1.json`.

Network / surface	Identifier
Ethereum Sepolia	chainId 11155111
Attestcoin source chain key	1
Creditcoin testnet	chainId 102031
Current factory	0x494490bBF748e59a659227F46510535BF3818442
BlockProver precompile	0x00FD2
ChainInfo precompile	0x00FD3

Appendix B. References

- Fair Witness repository: <https://github.com/Ay-obami/fair-witness>
- Fair Witness live application: <https://fair-witness.vercel.app/>
- Attestcoin Protocol: <https://attestcoin.org/>
- Attestcoin developer documentation: <https://docs.attestcoin.org/>
- docs/ATTESTCOIN_INTEGRATION.md
- docs/ARCHITECTURE.md
- docs/architecture/SECURITY_MODEL.md
- docs/architecture/POLICY_MODEL.md
- docs/architecture/PROPOSAL_SCHEMA.md
- docs/architecture/STRATEGY_MODEL.md
- docs/ADVERSARIAL_TEST_MATRIX.md
- docs/CONTROLLED_DEMO_RUNBOOK.md